

The AEC Framework

Advanced Reference — Quantitative Depth

Prepared for Pac Velz · AEC Mentorship Program

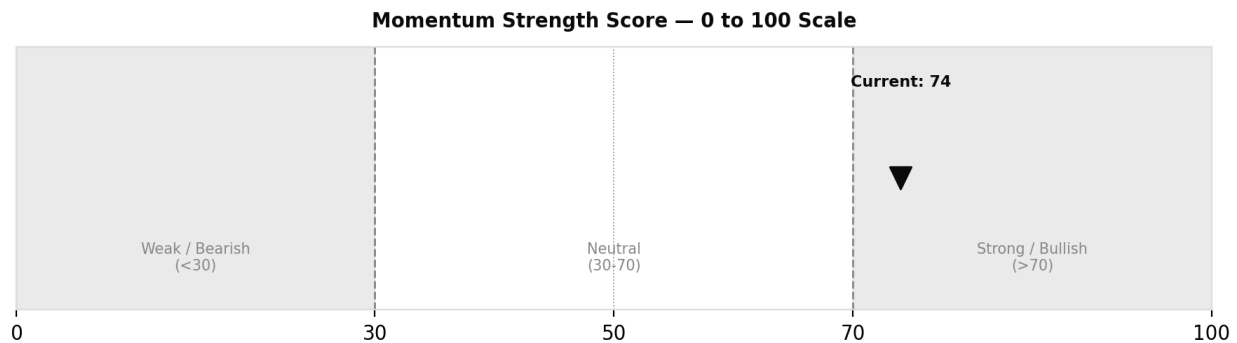
The math and mechanics behind the three-layer framework, condensed. No filler — formulas, worked examples, and the exact rules the system runs on.

I. ADVANCED MOMENTUM SCORING

Momentum strength is scored 0-100, built the same way as an RSI:

$$\text{RS} = \text{Average Gain (over N periods)} / \text{Average Loss (over N periods)}$$

$$\text{Momentum Score} = 100 - (100 / (1 + \text{RS}))$$



Above 70 confirms bullish momentum, below 30 confirms bearish. The 30-70 zone is neutral — no directional signal from Layer I regardless of how price looks visually.

Multi-Timeframe Weighting

HTF dominates the read: $\text{Weighted Momentum} = (\text{HTF Score} \times 0.65) + (\text{LTF Score} \times 0.35)$. Trade only when the weighted result exceeds 60 in either direction.

II. ADVANCED STATISTICAL DEVIATION

Core Z-score formula — deviation from the rolling mean in standard deviations:

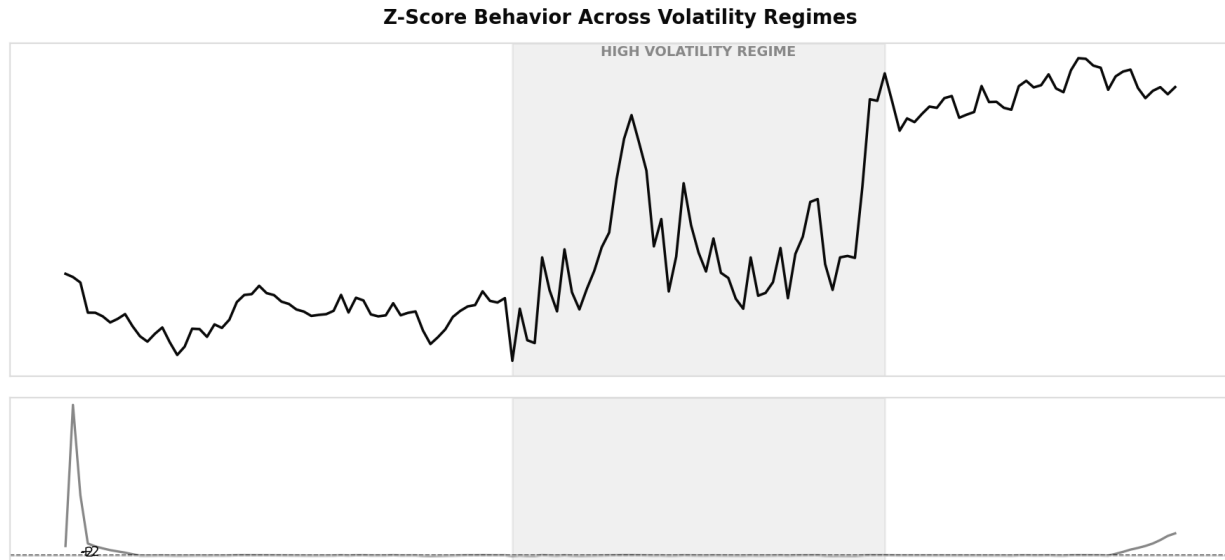
$$Z = (\text{Price} - \text{Rolling Mean}) / \text{Rolling Standard Deviation}$$

$$\text{Rolling Mean} = \text{Sum}(\text{Price}, N \text{ periods}) / N$$

$$\text{Rolling Std Dev} = \sqrt{(\text{Sum}((\text{Price} - \text{Mean})^2, N \text{ periods}) / N)}$$

Volatility Regime Adjustment

A fixed Z-score threshold behaves differently across volatility regimes — the same Z of 2.0 can be minor in a volatile stretch and a major outlier in a calm one. Fix: track rolling std dev over time; when it's meaningfully above its own longer-term average, widen thresholds (e.g. $\pm 2.0 \rightarrow \pm 2.5$) to avoid over-triggering.



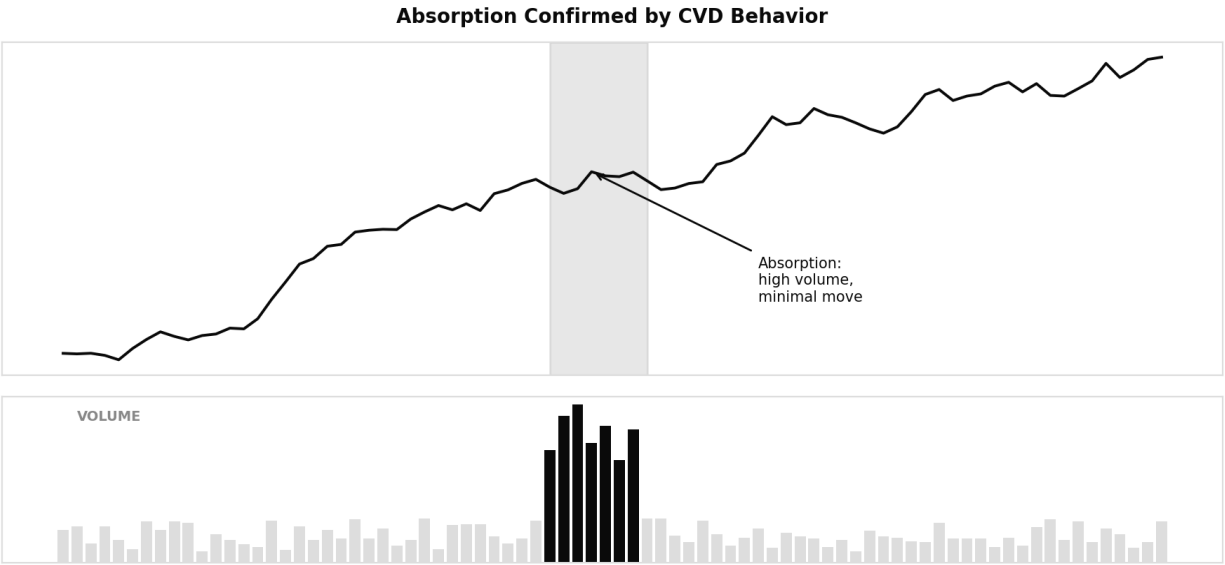
Shaded region = higher-volatility stretch. Z-score swings harder through it — without a regime adjustment, this window alone generates far more (lower-quality) signals than the calmer periods around it.

III. ADVANCED ORDER FLOW MECHANICS

Cumulative Volume Delta (CVD)

Running total of aggressive buy volume minus aggressive sell volume. The key metric for spotting divergence between price and what's actually happening underneath it.

Delta (per bar) = Aggressive Buy Volume – Aggressive Sell Volume
CVD = Running Sum of Delta across the session



Absorption, quantified: volume in the shaded window runs 3-4x the surrounding average while price barely moves — a large resting order absorbing flow without breaking.

Institutional Footprint Score

Three order-flow signals combined into one composite score (0-100):

Large Orders Present	Up to 45 points — scaled by count and relative size of resting orders vs. book average
CVD Trend Strength	Up to 30 points — scaled by magnitude and consistency of CVD direction
Absorption Confirmed	25 points — binary, awarded only when volume and price-movement criteria are both met

60+ required before Layer III is confirmatory in the aggregation logic below.

IV. POSITION SIZING & RISK MATH

Kelly Criterion (Fractional)

Position size derived from your actual edge — fractional Kelly avoids full-Kelly's excessive volatility:

$$\text{Kelly \%} = \text{Win Rate} - [(1 - \text{Win Rate}) / (\text{Avg Win} / \text{Avg Loss})]$$
$$\text{Position Size} = \text{Account Balance} \times (\text{Kelly \%} \times 0.5) - \text{half-Kelly, standard practice}$$

Worked example: 78% win rate, average win \$5,428, average loss \$3,000, \$100,000 account.

$\text{Kelly \%} = 0.78 - [(0.22) / (5428/3000)] = 0.78 - 0.122 = 0.658 \rightarrow 65.8\%$

Half-Kelly position size = $\$100,000 \times 0.329 = \$32,920$

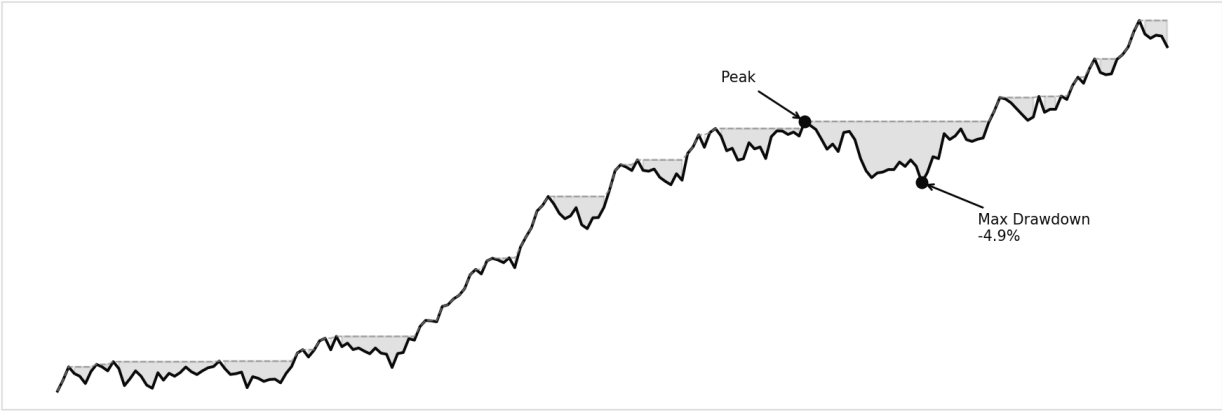
Even at half-Kelly, 65.8% raw Kelly produces a large recommended position — why most professional traders cap size well below what Kelly technically allows.

Risk of Ruin

Approximate probability of catastrophic account loss given your edge and sizing. A positive edge doesn't guarantee survival if sizing is too aggressive relative to strategy variance. Should trend toward zero as risk-per-trade drops — if it isn't near zero at your current sizing, your size is too large regardless of win rate.

V. PERFORMANCE ANALYTICS

Equity Curve — Peak-to-Trough Drawdown

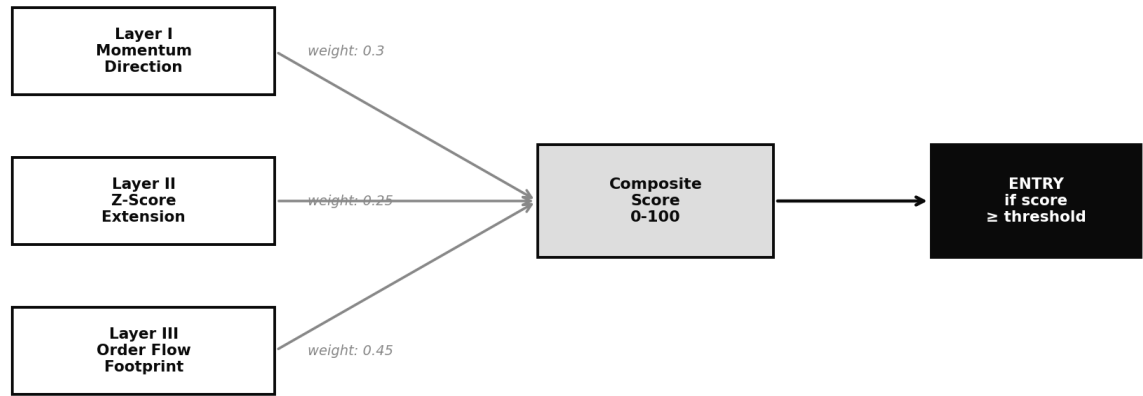


Max drawdown is tracked continuously, not reviewed after the fact. Strong win rate doesn't offset bad drawdown if sizing is miscalibrated.

Sharpe Ratio	$\text{Mean}(\text{Excess Returns}) / \text{StdDev}(\text{Excess Returns}) \times \sqrt{(\text{periods per year})}$ — risk-adjusted return
Expectancy	Sum of all trade P&L / Number of trades — the single number that determines long-run viability
Win Rate	Winning trades / Total trades $\times 100$ — meaningful only alongside expectancy, never alone

VI. FULL SIGNAL AGGREGATION LOGIC

How the three layers combine into one weighted decision:



Layer I Weight	0.30 — directional confirmation, necessary but not sufficient alone
Layer II Weight	0.25 — acts primarily as a filter/veto rather than a positive contributor
Layer III Weight	0.45 — weighted highest, since it provides the most precise, real-time confirmation
Entry Threshold	Composite score must reach 70+ before an entry is generated
Hard Veto	Regardless of composite score, Layer II failing the extension filter blocks entry entirely

VII. REQUIRED PLATFORMS

Three platforms, standard setup: one for charting, one for execution, one as full-service broker. Set up all three before your first session.

1. TradingView

Primary charting/analysis platform — where you'll mark Layer I structure and Layer II bands. Free tier works to start; paid tiers (\$0–\$60/mo) add indicators and layouts. Real-time data for most exchanges needs a separate subscription — budget for that past paper trading. Connects directly to Tradovate and IBKR.

2. Tradovate

Primary futures execution platform — cloud-based, where you'll trade NQ/ES live. Free Starter plan, no minimum balance. Monthly Member (\$99/mo) unlocks full API access for automation later. Micro contract margins as low as \$50 — reasonable place to start with small real size.

3. Interactive Brokers (IBKR)

Full-service broker — futures, equities, options. Same platform AEC uses for account management. Per-trade commissions, no platform fee. Connects to TradingView. Not needed day one, but part of the standard setup as your trading expands.

TradingView	tradingview.com — free tier to start
Tradovate	tradovate.com — free Starter plan, no minimum balance
Interactive Brokers	interactivebrokers.com — account setup required before funding

Platform fees, data, and commissions are separate from mentorship cost, paid directly to each provider. Pricing reflects 2026 public rates, subject to change — confirm current pricing directly on each site.

WHAT TO EXPECT WHEN YOU LOG IN

Every platform follows the same core layout regardless of branding. Diagrams below are illustrative generic mockups — not screenshots of any specific platform.

Generic Charting Workspace Layout — What to Expect



Core workspace: symbol/timeframe controls on top, drawing tools left, chart center, DOM/ladder right, order entry bottom. TradingView, Tradovate, and most platforms follow some version of this.

Generic Order Ticket — What Every Trade Requires

ORDER TICKET — NQ	
Order Type	Market / Limit / Stop
Quantity	1 contract
Entry Price	18,452.25
Stop Loss	18,438.00
Take Profit	18,480.75
Time in Force	Day
CONFIRM BUY CANCEL	

Every order needs the same core fields: type, quantity, entry, stop, target. Get these right before any live order.

Generic Account Dashboard — What You’ll Monitor Daily

ACCOUNT OVERVIEW					
Balance		Open P&L		Realized P&L Today	
\$102,448.00		+\$1,204.50		+\$842.00	
				Margin Used	
				\$4,200.00	
OPEN POSITIONS					
Symbol	Side	Qty	Entry	Current	P&L
NQZ6	LONG	1	18,452.25	18,461.50	+\$185.00

Balance, open/realized P&L, open positions — check this constantly during active hours.

A NOTE ON RISK

None of these formulas guarantee an outcome. They replace guessing with consistency, not risk with safety. All trading involves substantial risk of loss, including total loss of account balance. Every sizing formula assumes past win rate and average win/loss remain representative — never guaranteed. Charts are illustrative examples, not live market data.

This material is provided as part of AEC's mentorship program for educational purposes only and does not constitute investment advice. Past performance is not indicative of future results. All trading involves risk of loss.

Ascenda Enterprise Capital LLC · izaiah@ascendaenterprisecapital.com